- Data cleaning in easier terms makes raw data usable, and useful.
- it leads to easier visualization.

Steps

- import Data, creating a new database, use table data insert wizard
    - as of now we follow the insertion of data and just clicking next, side note you might want to change the date column into the proper but for now we will stick to how MySQL automatically assigns data types.
- **Step 1**
    - Remove Duplicates, usually the first thing to do, especially if its needed or the dataset needs no duplicate
- **Step 2**
    - Standardize the data, fix spellings, and issues
- **Step 3**
    - Null/Blank values
        - See if you can populate, should or should'nt
- Step 4
    - remove columns and rows that aren't needed
    - in real work place, have processes that automatically import data from different sources, if you remove column from the raw dataset then its a BIG NO
    - ***What should u do?***
      - create a type of staging table/raw dataset
        - copy the all the data from the raw data/table to the staging table
        - In my understanding, create a new table, copy all values in the raw into the new table.
    - *Why are we doing this?*
        - because we might be making mistakes all through out and we want to make the raw table available,
    - Best Practice
        - you should'nt work on the raw data, create a new table and do that one instead

REMOVING DUPLICATES

- note: its easier if there was a unique ID, for this there is none making it harder
- create a row_number for all partitioning by some columns (recommend all if not most)
    - what this does: it creates a row number for all, 1 if unique 2 or more if not.

    - Check/double check if there are errors.
- NOTE: CANNOT UPDATE/DELETE FROM A CTE IN MYSQL esentially delte is like update
- Since we cannot delete because of the CTE, we created another table instead adding `row_num` column and the data where the partitioning, rownum, Over, window functions were being used.
    - So we deleted the duplcated rows and now we delete the column (later)
    - ```
      SELECT *,
      ROW_NUMBER() OVER(
      PARTITION BY company, location, industry, total_laid_off, percentage_laid_off, `date`, stage, country, funds_raised_millions) AS row_num
      FROM layoffs_stage
      ```

      pasted is the data inserted.
    - ```
      -- DELETION
      DELETE
      FROM layoffs_stage2
      WHERE row_num > 1;

      INSERT INTO layoffs_stage2
      SELECT *,
      ROW_NUMBER() OVER(
      PARTITION BY company, location, industry, total_laid_off, percentage_laid_off, `date`, stage, country, funds_raised_millions) AS row_num
      FROM layoffs_stage
      ```

      insertion

STANDARDIZING DATA

- finding issues in the data and fixing it.
- First we Trimmed out spaces at the both ends of the company

```
SELECT company, TRIM(company)
FROM layoffs_stage2;

UPDATE layoffs_stage2
SET company = TRIM(company);
```

- Crypto
  Crypto Currency
  CryptoCurrency
    - This is an issue because in exploratory data naalysis, its their own rows and unique thing which is what WE DONT WANT. We want them all grouped together.
- ```
  UPDATE layoffs_stage2
  SET industry = 'Crypto'
  WHERE industry LIKE 'Crypto%'
  ```
    - Doing tthis makes it grouped into 1
  ```

- AT THIS POINT WE ARE JUST CHECKING DISTINCT FOR EVERY COLUMN MAKING SURE THAT THERE ARE NO ERRORS OR WHAT NOT
-
```sql
SELECT DISTINCT country, TRIM(TRAILING '.' FROM country)
FROM layoffs_stage2
ORDER BY 1;
```

United States	United States
United States.	United States

- This is pretty cool i never learned this.

NOTE: timeseries, date(text rn) should be changed into the correct data type

- how can we do that?
-
```sql
SELECT `date`,
STR_TO_DATE(date, '%m/%d/%Y')
FROM layoffs_stage2
;
```

date	STR_TO_DATE(date, '%m/%d/%Y')
12/16/2022	2022-12-16
7/25/2022	2022-07-25
11/17/2022	2022-11-17
1/27/2023	2023-01-27
7/13/2022	2022-07-13
1/10/2023	2023-01-10
8/4/2022	2022-08-04
4/2/2020	2020-04-02
6/1/2022	2022-06-01
9/1/2022	2022-09-01
7/28/2022	2022-07-28
8/29/2022	2022-08-29
6/3/2022	2022-06-03

- Here we can see how its done, The order is small m, small d, capital Y
  - if its all capitalized
    - NULL
  - if M is capitalized and d isnt

        - NULL
    - if d is capitalized and m isnt
        - NULL
    - if d and m is revered,
        - some null some not
- probbably due to how the data formats the date and how mysql formatting date.
- now were going to change the data type for `date` we always have to stay on the staging table and not the actual raw data
- Here is the code for changing data type
    -
    ```
    ALTER TABLE layoffs_stage2
    MODIFY COLUMN `date` DATE;
    ```

NULLS/BLANKS

- this is normal, there will be nulls and blanks
    - we need to think about what we should do.
        - null or populate or whatever
- Here at this point we are self joining the table and finding out the industry because of the missing values, now we cannot just put randomly a industry because it may be different
    -
    ```
    SELECT t1.company, t2.company, t1.industry, t2.industry
    FROM layoffs_stage3 t1
    JOIN layoffs_stage3 t2
        ON t1.company = t2.company
    WHERE (t1.industry IS NULL OR t1.industry = '') AND t2.industry IS NOT NULL;
    ```

    Result Grid | Filter Rows: | Export: | Wrap Cell Content:

company	company	industry	industry
Airbnb	Airbnb		
Airbnb	Airbnb		Travel
Carvana	Carvana		
Carvana	Carvana		Transportation
Carvana	Carvana		Transportation
Juul	Juul		
Juul	Juul		Consumer

    -
        - Here is an example, showing null on industry but on the 2nd result its travel, transportation, and consumer, we can assume that the industry for those are those respectively.
    - Now here we encountered a problem, maybe because its blank and not NULL?

```sql
UPDATE layoffs_stage3 t1
JOIN layoffs_stage3 t2
    ON t1.company = t2.company
SET t1.industry = t2.industry
WHERE (t1.industry IS NULL OR t1.industry = '') AND t2.industry IS NOT NULL;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

company	company	industry	industry
Airbnb	Airbnb		
Airbnb	Airbnb		Travel
Carvana	Carvana		
Carvana	Carvana		Transportation
Carvana	Carvana		Transportation
Juul	Juul		
Juul	Juul		Consumer

-
    - update is not working
- Make sure that u update ONLY what u can and from information from your datasets.
- IF YOU DONT NEED THE INFRMATION OR CANNOT GET ANYTHING FROM IT DELETE
-

SUMMARY

- Remove duplicates
- standardize data
- Null/blanks
- remove any columns or rows that are not needed